A Music Theory

A plain-language guide to notes, guitar, bass, ukulele and chords, written for someone who knows nothing about music.

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1. The 12 notes

A **note** is a named pitch: how high or low a sound is. Music uses only **12 different notes**, and then they repeat higher and lower. Think of the 12 hours on a clock: after 12 comes 1 again.

A	A♯	B	C	C♯	D	D♯	E	F	F♯	G	G♯

- The letters run **A to G**, with extra in-between notes written with ♯ ("sharp": a tiny step higher).
- □ ("flat") means a tiny step lower. A♯ and B♭ are the **same sound** with two names; likewise C♯ = D♭, D♯ = E♭, F♯ = G♭, G♯ = A♭. This guide uses sharps throughout.
- There is no in-between note from B to C, or from E to F. That is just how it worked out.
- A **step** (one fret on a guitar, one neighbouring key on a piano) moves you to the next note in the list.

Key idea: everything in music, on any instrument, uses these same 12 notes.

2. Guitar fretboard

- A guitar has **6 strings**, tuned (thickest to thinnest) to **E, A, D, G, B, E**.
- **Frets** are the metal bars along the neck. Pressing a string at a fret shortens it, so the sound gets higher.
- **Each fret = one step up** the list of 12 notes. After **12 frets** you are back at the same note, one cycle higher (the double dot on many guitars).
- Neighbouring strings are **5 frets apart**, except G to B, which is 4. That is why chord shapes bend slightly around the B string.
- The same note appears in **several places** on the neck. The guitar has the same 12 notes repeated.

All notes, frets 0 to 21. Columns run from the thinnest string (1st) to the thickest (6th). The line under fret 12 shows where the pattern starts over.

Fret	E (1st, thin)	B (2nd)	G (3rd)	D (4th)	A (5th)	E (6th, thick)
0 (open)	E	B	G	D	A	E
1	F	C	G♯	D♯	A♯	F
2	F♯	C♯	A	E	B	F♯
3	G	D	A♯	F	C	G
4	G♯	D♯	B	F♯	C♯	G♯
5	A	E	C	G	D	A
6	A♯	F	C♯	G♯	D♯	A♯
7	B	F♯	D	A	E	B
8	C	G	D♯	A♯	F	C
9	C♯	G♯	E	B	F♯	C♯
10	D	A	F	C	G	D
11	D♯	A♯	F♯	C♯	G♯	D♯
12	E	B	G	D	A	E
13	F	C	G♯	D♯	A♯	F
14	F♯	C♯	A	E	B	F♯
15	G	D	A♯	F	C	G
16	G♯	D♯	B	F♯	C♯	G♯
17	A	E	C	G	D	A
18	A♯	F	C♯	G♯	D♯	A♯
19	B	F♯	D	A	E	B
20	C	G	D♯	A♯	F	C
21	C♯	G♯	E	B	F♯	C♯

3. Bass guitar fretboard

- Same 12 notes. A standard bass has **4 strings**: **E, A, D, G**, thickest to thinnest.
- These are the same names as the four thickest guitar strings, but they sound **one full cycle (12 frets) lower**.
- Every string is **5 frets** from its neighbour, so the pattern is completely regular.
- The bass usually plays **one note at a time** rather than full chords.

All notes, frets 0 to 21 (thinnest string first).

Fret	G (thinnest)	D	A	E (thickest)
0 (open)	G	D	A	E
1	G♯	D♯	A♯	F
2	A	E	B	F♯
3	A♯	F	C	G
4	B	F♯	C♯	G♯
5	C	G	D	A
6	C♯	G♯	D♯	A♯
7	D	A	E	B
8	D♯	A♯	F	C
9	E	B	F♯	C♯
10	F	C	G	D
11	F♯	C♯	G♯	D♯
12	G	D	A	E
13	G♯	D♯	A♯	F
14	A	E	B	F♯
15	A♯	F	C	G
16	B	F♯	C♯	G♯
17	C	G	D	A
18	C♯	G♯	D♯	A♯
19	D	A	E	B
20	D♯	A♯	F	C
21	E	B	F♯	C♯

4. Ukulele fretboard

- A standard ukulele has **4 strings** tuned **G, C, E, A**, listed from the string nearest your face to the one nearest the floor.
- **The G string sounds higher than the C string** next to it, so it is not the lowest string. This is called re-entrant tuning and gives the ukulele its bright, jingly sound.
- Ordered by pitch, low to high, the strings are **C, E, G, A**.
- Putting a guitar capo on fret 5 turns the four thinnest guitar strings (D, G, B, E) into the ukulele's G, C, E, A.
- Ukuleles usually have 12 to 19 frets. The table shows 0 to 15.

All notes, frets 0 to 15, in the order of the strings on the instrument.

Fret	G (4th)	C (3rd)	E (2nd)	A (1st)
0 (open)	G	C	E	A
1	G♯	C♯	F	A♯
2	A	D	F♯	B
3	A♯	D♯	G	C
4	B	E	G♯	C♯
5	C	F	A	D
6	C♯	F♯	A♯	D♯
7	D	G	B	E
8	D♯	G♯	C	F
9	E	A	C♯	F♯
10	F	A♯	D	G
11	F♯	B	D♯	G♯
12	G	C	E	A
13	G♯	C♯	F	A♯
14	A	D	F♯	B
15	A♯	D♯	G	C

5. What is a chord?

A **chord** is **several notes played at the same time**. One note is like one singer; a chord is like a small choir.

- Most chords use **3 notes**, sometimes 4 or more.
- A chord is built from a **starting note** (the root, which gives the chord its name) plus other notes chosen by a fixed pattern of steps.
- Example: the **C chord** is the notes **C, E and G** played together.
- On guitar and ukulele you press certain frets with your fingers and strum, so the notes ring together. Each chord has a learnable "shape".
- Songs are mostly a **sequence of chords** under a melody. Many pop songs use only 3 or 4 chords.
- A letter alone means major (**C**); a letter with a small m means minor (**Cm**).

6. Major and minor triads

A **triad** is a chord of exactly **3 notes**: the **root** (the name note), the **third** (decides the mood) and the **fifth** (always 7 steps above the root, adds stability).

	Root → third	Third → fifth	Mood
Major	4 steps	3 steps	Bright, happy
Minor	3 steps	4 steps	Dark, sad

- **C major:** C → 4 steps → E → 3 steps → G = **C, E, G**
- **C minor:** C → 3 steps → D♯ → 4 steps → G = **C, D♯, G**
- **A major:** A, C♯, E **A minor:** A, C, E

Key idea: moving only the middle note by one step turns bright into sad.

7. Every chord type

All examples start on **C**. The "steps" are counted up from the root (step 0), one step = one fret. To make the chord from any other root, count the same steps from that note. For example, G7 uses steps 4, 7, 10 from G: G, B, D, F.

The 2-note chord

Name	Steps from root	Notes in C	Sound
5 (C5), "power chord"	7	C, G	Strong and hollow, neither happy nor sad. Very common in loud rock guitar.

The 3-note chords

Name	Steps from root	Notes in C	Sound
Major (C)	4, 7	C, E, G	Bright, happy
Minor (Cm)	3, 7	C, D♯, G	Dark, sad
Diminished (Cdim)	3, 6	C, D♯, F♯	Tense, spooky, wants to move on
Augmented (Caug)	4, 8	C, E, G♯	Dreamy, unsettled
Sus2 (Csus2)	2, 7	C, D, G	Open and airy, neither happy nor sad
Sus4 (Csus4)	5, 7	C, F, G	Open, hanging, wants to fall back to major

"Sus" is short for "suspended": the middle note is swapped for a neighbour, so the chord is neither happy nor sad and feels like it is hanging.

The 4-note chords

Name	Steps from root	Notes in C	Sound
7 (C7)	4, 7, 10	C, E, G, A♯	Bluesy, restless, wants to go somewhere
Maj7 (Cmaj7)	4, 7, 11	C, E, G, B	Smooth, dreamy, relaxed
m7 (Cm7)	3, 7, 10	C, D♯, G, A♯	Mellow, soft sad, jazzy
7sus4 (C7sus4)	5, 7, 10	C, F, G, A♯	Open and floating with a bluesy edge
Dim7 (Cdim7)	3, 6, 9	C, D♯, F♯, A	Very tense and dramatic (suspense music)

Ways to remember

- **7** = major chord + a note 10 steps up (bluesy). **maj7** = major chord + a note 11 steps up (smooth). **m7** = minor chord + the same 10-step note.
- **dim** squeezes the chord (3+3 steps); **dim7** adds one more equal step (3+3+3), so it sounds very unstable. **aug** stretches it (4+4).
- **sus** swaps the middle note, removing the happy/sad mood.

8. One-page cheat sheet

The 12 notes	A A♯ B C C♯ D D♯ E F F♯ G G♯ (then A again)
One fret	One step up the list. 12 frets = back to the same note, higher.
Guitar	6 strings: E A D G B E (thick to thin). Strings are 5 frets apart, except G to B (4).
Bass	4 strings: E A D G. All 5 frets apart. Sounds one cycle lower than guitar.
Ukulele	4 strings: G C E A. The G is higher than the C (re-entrant tuning).
Chord	3 or more notes played together. Letter = start note, suffix = mood.
Major	root, +4, +7 (bright)
Minor	root, +3, +7 (sad)
5	root, +7 (power chord)
7	root, +4, +7, +10 (bluesy)
Maj7	root, +4, +7, +11 (smooth)
m7	root, +3, +7, +10 (mellow)
Dim / Dim7	root, +3, +6 / root, +3, +6, +9 (tense)
Aug	root, +4, +8 (dreamy)
Sus2 / Sus4	root, +2, +7 / root, +5, +7 (hanging)
7sus4	root, +5, +7, +10 (floating, bluesy)

The tip to take away: learn the open string names of your instrument and count up one fret at a time, and learn each chord as a recipe of steps up from its root. Then you can work out any note or chord yourself.